Algebraic Formulation of the $3n + 5$ Collatz Orbit in Jacobsthal Space: The $3J + 5J(0,1)$ Transformation

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Abstract

In previous studies, we algebraically formulated the Collatz conjecture (the $3n + 1$ problem) as an affine transformation $3J + J(0,1)$ within Jacobsthal (J) space. This study aims to generalize this algebraic framework and apply it to the orbital analysis of the "$3n + 5$ problem," a dynamical system governed by a different rule set. By defining the following affine transformation for a spinor $J(a, b)$ with an internal state $a = (b + 5)/2$, we isolated the additive noise "+5" to verify the dynamics of pure scaling:

$$\mathbf{3}\boldsymbol{J}(\boldsymbol{a}, \boldsymbol{b}) + \mathbf{5}\boldsymbol{J}(\mathbf{0}, \mathbf{1})$$

Through this method, we confirmed that the known loops in the $3n + 5$ space $(1, 5, 19, 23, 187, 347)$ can be uniformly described under the common framework of J-Spinor dynamics.

1. Introduction

The standard Collatz operation ($3n + 1$ multiplication and division by 2) ultimately converges to a trivial loop ($1 \to 4 \to 2 \to 1$). In our previous research, we defined an odd node b as a two-component column vector (J-Spinor) $J(a, b)$ in Jacobsthal (J) space, representing the Collatz operation as a vector transformation $3J(a, b) + J(0,1)$. Conversely, in a dynamical system where the scalar operation is modified to $n \mapsto 3n + 5$ (the $3n + 5$ problem), a complex orbital space is formed where multiple independent cycles $(1, 5, 19, 23, 187, 347)$ coexist instead of a single loop. In this study, we investigated whether the algebraic methods established in the $3n + 1$ space—namely, "J-space 2-adic reduction (J-Reduction)" and "J-Normalization"—function effectively in a space with a different additive noise, such as $3n + 5$.

2. Vector Definition of J-Spinor in $3n + 5$ Space and The $3J + 5J(0,1)$ Transformation

We define the odd node b and its internal state a in the $3n + 5$ space as follows:

$$a = (b + 5)/2$$

$$J(a, b) = \begin{pmatrix} a \\ b \end{pmatrix}$$

For this spinor, which satisfies the fundamental relation $b = 2a - 5$, we define the Collatz operation in $3n + 5$ as the following affine transformation:

$$3\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 3a \\ 3b + 5 \end{pmatrix}$$

Through this transformation, the additive noise "+5" is isolated strictly in the lower b component, allowing the upper a component to act as a pure 3-fold scaling component unaffected by the noise.

3. J-Reduction and J-Normalization

In Collatz orbits, groups (branches) of odd nodes converging to the same odd node form an equivalence class. Here, we define a set of J-sequences sharing the identical **Reduced Spinor** as a "**J equivalence class**." For instance, by applying the $3n + 5$ operation and 2-adic reduction, the odd terms of the following $J(4,3)$ sequence all arrive at the common odd node 7:

$$J(4,3) = \{4, 3, 11, 17, 39, 73, \dots\}$$

$$J(4,3)_{odd} = \{3, 17, 73, \dots\}$$

However, fractional states arise during division-by-2 shifts where $k \geq 2$. For example, at odd node 17, the standard operation yields $3(17) + 5 = 56 = 7 \times 2^3$ $(k = 3)$. If directly computed using J-spinors, fractions appear: $J(11,17) \rightarrow J(33,56) \rightarrow J(33/8,7)$. To prevent the introduction of fractions, rather than continuing the calculation as-is, one must re-select the Reduced Spinor located at the root of this branch as the representative element. We term this process J-space 2-adic reduction (J-Reduction).

For example, the Reduced Spinor of the $J(11,17)$ sequence is $J(4,3)$. At this odd node 3, the operation is $3(3) + 5 = 14 = 7 \times 2^1$ $(k = 1)$, meaning the J-spinor calculation proceeds as $J(4,3) \rightarrow J(12,14) \rightarrow J(6,7)$, successfully maintaining integer states.

Next, we consider cases where $k \geq 2$ still occurs even after applying the $3n + 5$ operation to a Reduced Spinor.

For instance, in the case of $J(6,17)$, at odd node 17, the calculation $3(17) + 5 = 56 = 7 \times 2^3$ $(k = 3)$ leads to the spinor transition $J(6,17) \rightarrow J(18,56) \rightarrow J(9/4,7)$.

$$J(9/4,7) = \{9/4, 7, 23/2, 51/2, \dots\}$$

By multiplying this sequence by 2, it can be normalized as a J-sequence (J-Normalization) as follows:

$$J(14,23) = \{14, 23, 51, 97, \dots\}$$

4. J-Space Expansion of Representative Loops

We dissect the multiple loop structures present in the $3n+5$ space $(1, 5, 19, 23, 187, 347)$ through J-Expansion (definition of standard spinor → extraction of reduced spinor → transformation and normalization).

4.1 Loop 1 $(1 \to 1)$

Loop 1 completes a cycle in one step $(1 \to 1,\ k = 3)$.

- Standard Spinor: From $b = 1,\ a = (1+5)/2 = 3 \Longrightarrow J(3,1)$

In J-Expansion, this naturally extends into the negative odd domain as shown below.

- J-Expansion: $J = \{2, -1, 3, 1, 7, 9, 23, \dots\}$
- Reduced Spinor: $J(2,-1)$

$$3J(2,-1) + 5J(0,1) = J(6,2) = 2J(3,1)$$

- Next Spinor: $J(3,1)$

As a result of factoring out the scalar 2, the orbit returns to the original $J(3,1)$ through a shift down $(1/2$ scaling).

4.2 Loop 5 $(5 \to 5)$

Loop 5 completes a cycle in one step $(5 \to 5,\ k = 2)$.

- Standard Spinor: From $b = 5,\ a = (5+5)/2 = 5 \Longrightarrow J(5,5)$
- J-Expansion: $J = \{0, 5, 5, 15, 25, 55, \dots\}$
- Reduced Spinor: $J(0,5)$

Applying the transformation to the extracted fundamental representative $J(0,5)$:

$$3J(0,5) + 5J(0,1) = \begin{pmatrix} 0 \\ 15 \end{pmatrix} + \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ 20 \end{pmatrix} = 4\begin{pmatrix} 0 \\ 5 \end{pmatrix} = 4J(0,5)$$

- Next Spinor: $J(0,5)$

The orbit returns to the original $J(0,5)$ through a shift down $(1/4$ scaling). Notably, only the orbit 5 loop acts as an exact 5-fold scaling $(5J(0,1))$ of the trivial loop (orbit 1) in the $3n+1$ space, thereby preserving the structure of the original space. The remaining loops $(1, 19, 23, 187, 347)$ may be interpreted as supplementary closed orbits that appear when the additive noise is changed from $+1$ to $+5$.

4.3 Loop 19 $(19 \to 31 \to 49 \to 19)$

The orbit of 19 returns to its starting point after three steps. Here, we analyze the final return step $49 \to 19$ $(k = 3)$.

- Standard Spinor: From $b = 49,\ a = (49 + 5)/2 = 27 \Longrightarrow J(27,49)$
- J-Expansion: $J = \{8, 11, 27, 49, \dots\}$
- Reduced Spinor: $J(8,11)$
- $3J + 5J(0,1)$ Transformation:

$$3\begin{pmatrix} 8 \\ 11 \end{pmatrix} + \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 24 \\ 38 \end{pmatrix} \xrightarrow{\div 2^1} \begin{pmatrix} 12 \\ 19 \end{pmatrix}$$

- Next Spinor: $J(12,19)$

The $J(12,19)$ reached via a shift down $(1/2$ scaling) matches the standard spinor of 19 $(a = 12)$.

4.4 Loop 23 $(23 \to 37 \to 29 \to 23)$

The orbit of 23 returns to its starting point after three steps. Here, we analyze the final return step $29 \to 23$ $(k = 2)$.

- Standard Spinor: From $b = 29,\ a = (29 + 5)/2 = 17 \Longrightarrow J(17,29)$
- J-Expansion: $J = \{6, 17, 29, \dots\}$
- Reduced Spinor: $J(6,17)$
- $3J + 5J(0,1)$ Transformation:

$$3\begin{pmatrix} 6 \\ 17 \end{pmatrix} + \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 18 \\ 56 \end{pmatrix} \xrightarrow{\div 2^3} \begin{pmatrix} 9/4 \\ 7 \end{pmatrix}$$

- Intermediate Spinor: $J(9/4,7)$
- J-Expansion: $J = \{9/4, 7, 23/2, \dots\}$
- J-Normalization: $J = \{9/2, 14, 23, \dots\}$
- Next Spinor: $J(14,23)$

After a shift down $(1/8$ scaling), J-Normalization is applied to convert $\begin{pmatrix} 9/4 \\ 7 \end{pmatrix}$ into an integer-valued spinor. Consequently, the resulting $J(14,23)$ matches the standard spinor of 23.

4.5 Loop 187 $(187 \to 283 \to \cdots \to 1993 \to 187)$

The orbit of 187 returns to its starting point after 17 steps. Here, we analyze the final return step $1993 \to 187$ $(k = 5)$.

- Standard Spinor: From $b = 1993,\ a = 999 \Longrightarrow J(999,1993)$
- J-Expansion: $J = \{64, 123, 251, 497, 999, 1993, \ldots\}$
- Reduced Spinor: $J(64,123)$
- $3J + 5J(0,1)$ Transformation:

$$3\begin{pmatrix}64\\123\end{pmatrix} + \begin{pmatrix}0\\5\end{pmatrix} = \begin{pmatrix}192\\374\end{pmatrix} \xrightarrow{\div 2^1} \begin{pmatrix}96\\187\end{pmatrix}$$

- Next Spinor: $J(96,187)$

The $J(96,187)$ reached via a shift down $(1/2$ scaling) matches the standard spinor of 187.

4.6 Loop 347 $(347 \to 523 \to \cdots \to 461 \to 347)$

The orbit of 347 returns to its starting point after 17 steps. Here, we analyze the final return step $461 \to 347$ $(k = 2)$.

- Standard Spinor: From $b = 461,\ a = 233 \Longrightarrow J(233,461)$
- J-Expansion: $J = \{114, 233, 461, 927, 1849, \ldots\}$
- Reduced Spinor: $J(114,233)$
- $3J + 5J(0,1)$ Transformation:

$$3\begin{pmatrix}114\\233\end{pmatrix} + \begin{pmatrix}0\\5\end{pmatrix} = \begin{pmatrix}342\\704\end{pmatrix} \xrightarrow{\div 2^6} \begin{pmatrix}171/32\\11\end{pmatrix}$$

- Intermediate Spinor: $J(171/32,11)$
- J-Expansion: $J = \{171/32, 11, 347/16, \ldots\}$
- J-Normalization: $J = \{171/2, 176, 347, \ldots\}$
- Next Spinor: $J(176,347)$

After a shift down $(1/64$ scaling), J-Normalization is applied to convert $\begin{pmatrix}171/32\\11\end{pmatrix}$ into an integer-valued spinor. Consequently, the resulting $J(176,347)$ matches the standard spinor of 347. All J-spinor transitions in Loop 347 are summarized in **Table 1**.

Table 1. Collatz Orbit for Initial Node 347 in J-Spinor Representation under $3n+5$ Dynamics

Step	Odd Node	Standard J	Reduced J	3J + 5J(0,1)	Next J
0	347	J(176, 347)	J(176, 347)	J(528, 1046)	J(264, 523)
1	523	J(264, 523)	J(264, 523)	J(792, 1574)	J(396, 787)
2	787	J(396, 787)	J(396, 787)	J(1188, 2366)	J(594, 1183)
3	1183	J(594, 1183)	J(594, 1183)	J(1782, 3554)	J(891, 1777)
4	1777	J(891, 1777)	J(224, 443)	J(672, 1334)	J(336, 667)
5	667	J(336, 667)	J(336, 667)	J(1008, 2006)	J(504, 1003)
6	1003	J(504, 1003)	J(504, 1003)	J(1512, 3014)	J(756, 1507)
7	1507	J(756, 1507)	J(756, 1507)	J(2268, 4526)	J(1134, 2263)
8	2263	J(1134, 2263)	J(1134, 2263)	J(3402, 6794)	J(1701, 3397)
9	3397	J(1701, 3397)	J(848, 1701)	J(2544, 5108)	J(1277, 2549)
10	2549	J(1277, 2549)	J(636, 1277)	J(1908, 3836)	J(959, 1913)
11	1913	J(959, 1913)	J(118, 241)	J(354, 728)	J(182, 359)
12	359	J(182, 359)	J(182, 359)	J(546, 1082)	J(273, 541)
13	541	J(273, 541)	J(134, 273)	J(402, 824)	J(206, 407)
14	407	J(206, 407)	J(206, 407)	J(618, 1226)	J(309, 613)
15	613	J(309, 613)	J(152, 309)	J(456, 932)	J(233, 461)
16	461	J(233, 461)	J(114, 233)	J(342, 704)	J(176, 347)
17	347	J(176, 347)	-	-	LOOP DETECTED

The J-spinor transition process of the Collatz orbit from the initial node $b_0 = 347$ to its loop closure (node 347). Each column represents the following algebraic operations:

- **Odd Node:** The current odd node b in the standard $3n + 5$ Collatz orbit.
- **Standard J:** The standard spinor $J(a, b)$ defined by the internal state $a = (b + 5)/2$.
- **Reduced J:** The reduced spinor extracted via J-space 2-adic reduction, serving as the foundational input vector for the current step.
- **3J + 5J(0,1):** The result of applying the affine transformation to the reduced input spinor.
- **Next J:** The normalized standard spinor of the next odd node, obtained through sequence expansion and normalization without introducing fractional components.

5. Orbit Confluence in Inverse Tree Structure and Projection of $4b + 5$

In the $3n + 5$ space, multiple branches merge into specific nodes (e.g., node 19).

$$11 \to 19, \quad 49 \to 19, \quad 201 \to 19$$

Tracing the standard spinor of each node back to its root via J-Reduction reveals that all converge to a single reduced spinor $J(8,11)$. The expanded J-sequence of this $J(8,11)$ is as follows:

$$J(8,11) = \{8, \mathbf{11}, 27, \mathbf{49}, 103, \mathbf{201}, \dots\}$$

In other words, the orbit confluence observed in the inverse tree structure can be interpreted as a single expanded J-sequence emerging as a projection into multiple branches via the $4b + 5$ recurrence acting as a 2-adic amplification process.

6. Conclusion

In this study, we attempted to describe the Collatz operation in the $3n + 5$ space using the algebraic framework $3J + 5J(0,1)$.

1) By isolating the additive noise "$+5$" via an affine transformation, we explicitly described the multiplicative component of the orbital calculation.
2) We demonstrated that the six distinct loops within the space can be uniformly described under the common dynamics of the J-Spinor.
3) The numerous loop structures arising in the $3n + 5$ space may be interpreted as supplementary closed orbits that appear when the additive noise is changed from $+1$ to $+5$. Among them, it was suggested that only the "orbit 5 loop" functions as an exact 5-fold scaling ($5J(0,1)$) that directly inherits the structure of the $3n + 1$ space.
4) We showed that even for larger loops such as 187 and 347, analysis can be performed without changing the rules by extracting the "root" via reduction from the standard spinor of each node, and applying the normalization algorithm from there. These results suggest that the J-Spinor framework may have applicability beyond a single dynamical system ($3n + 1$), potentially extending to dynamical systems with arbitrary odd noise ($3n + m$).

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